

Pulse sequence for simple, decouple and manipulating Nuclear Overhauser Effect (NOE)

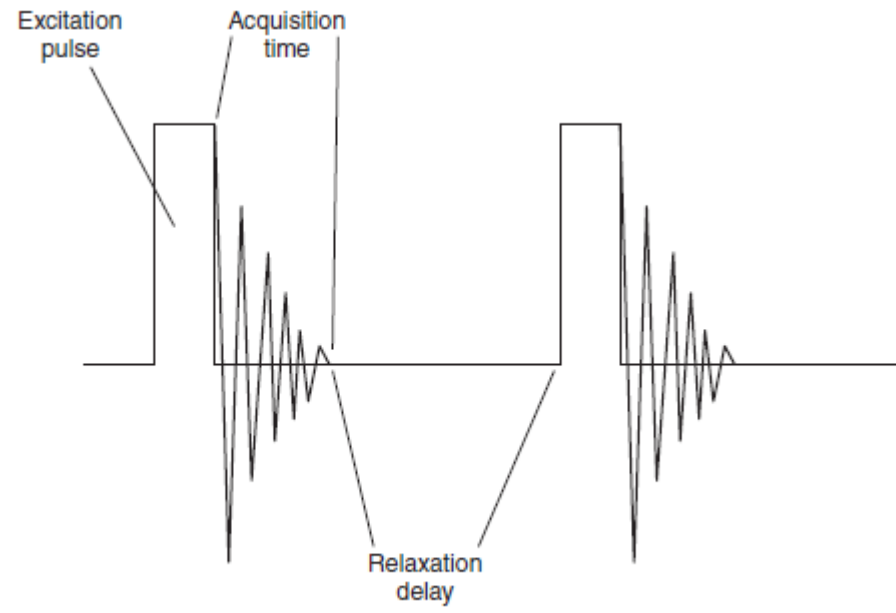


FIGURE 9.1 A simple pulse sequence.

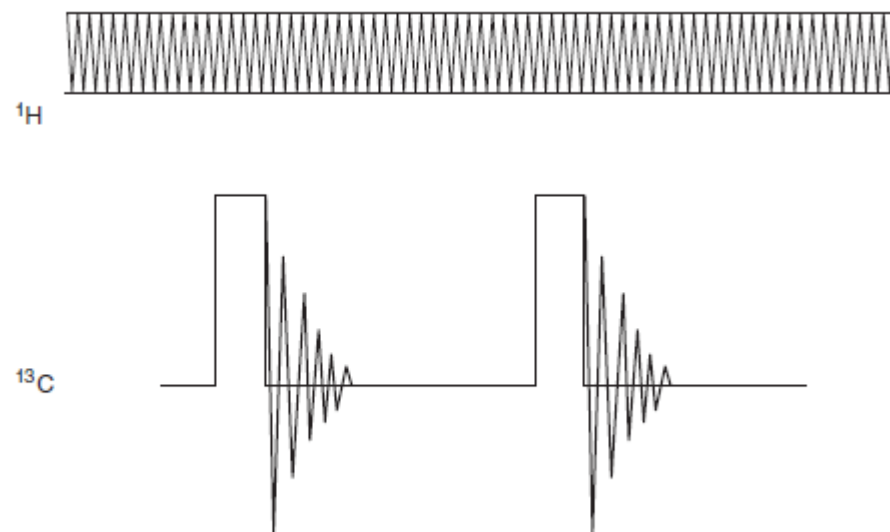


FIGURE 9.2 A proton-decoupled ^{13}C NMR pulse sequence.

NOE-enhanced proton-coupled spectrum or a gated decoupling spectrum

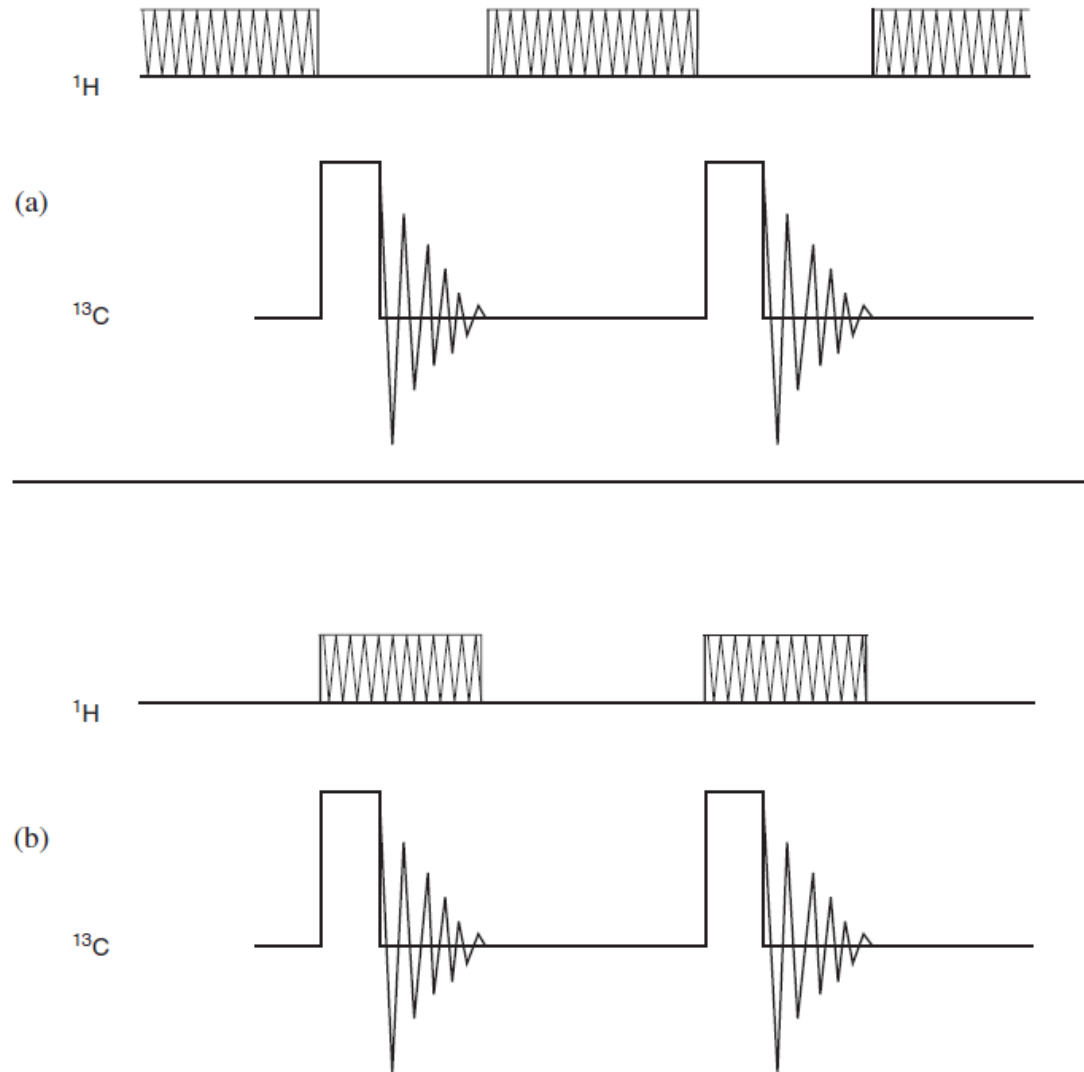


FIGURE 9.3 A simple pulse sequence. (a) A pulse sequence for gated decoupling; (b) a pulse sequence for inverse gated decoupling.

a) The effect of this pulse sequence is to allow the nuclear Overhauser effect to develop while the decoupler is on. Because the decoupler is switched off during the excitation pulse, spin decoupling of the ^{13}C atoms is not observed (a proton-coupled spectrum is seen). The nuclear Overhauser enhancement decays over a relatively long time period; therefore, most of the enhancement is retained as the FID is collected.

NOE-enhanced proton-coupled spectrum or a gated decoupling spectrum

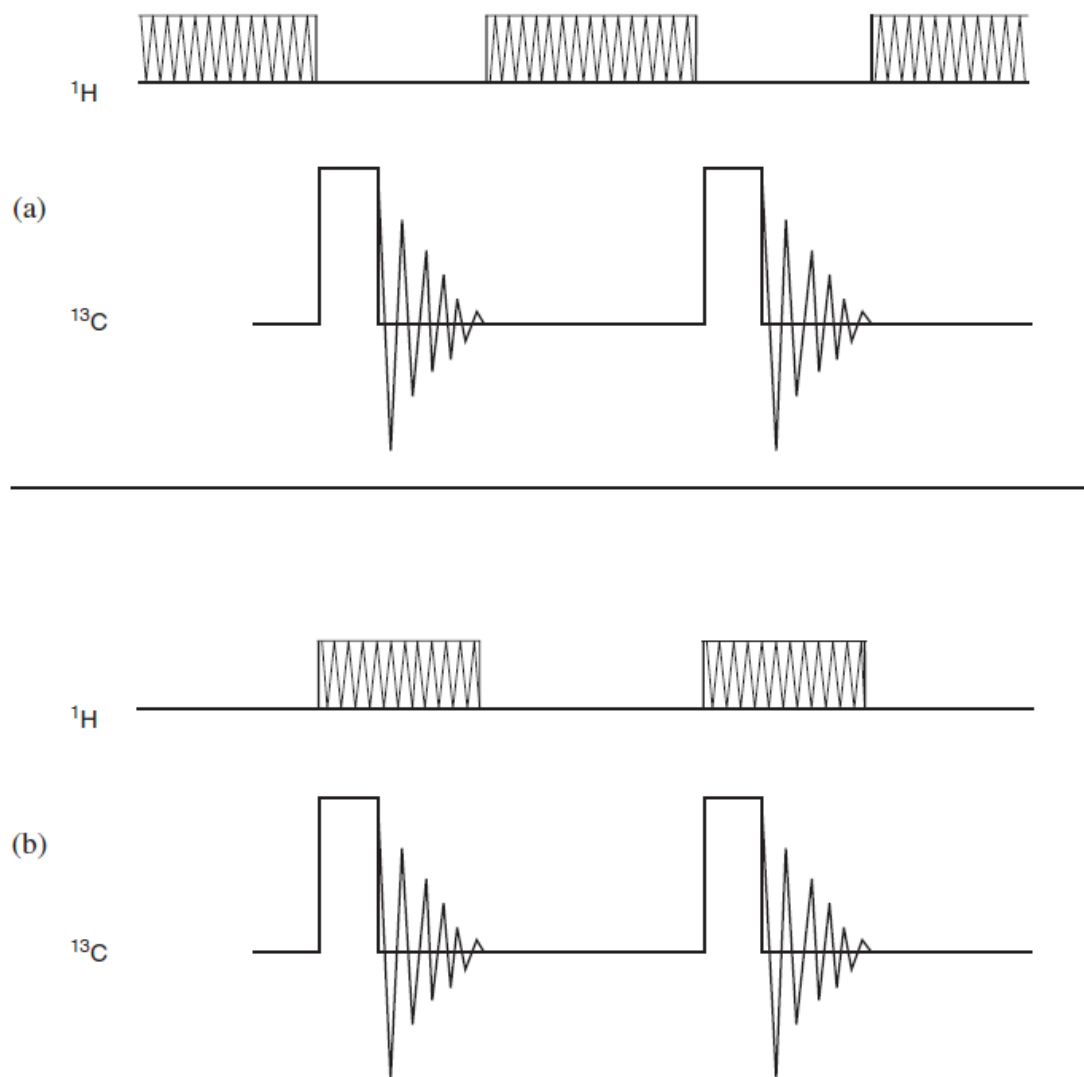


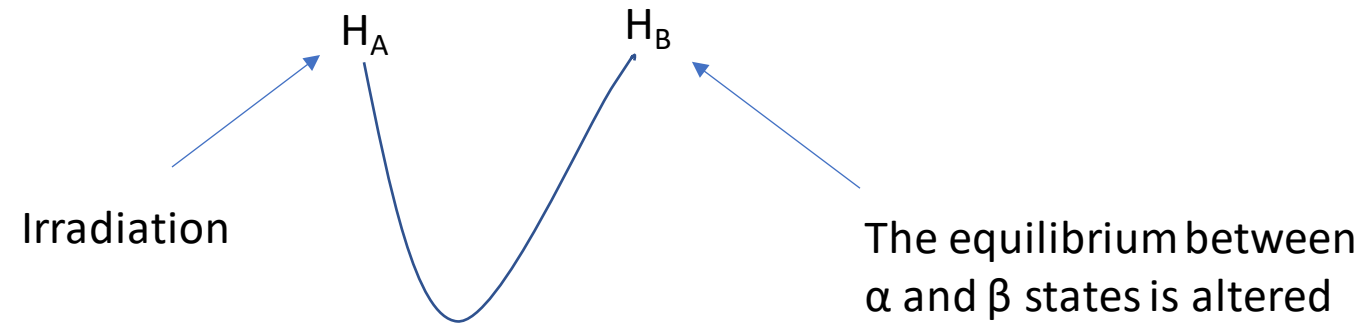
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a) The effect of this pulse sequence is to allow the nuclear Overhauser effect to develop while the decoupler is on. Because the decoupler is switched off during the excitation pulse, spin decoupling of the ^{13}C atoms is not observed (a proton-coupled spectrum is seen). The nuclear Overhauser enhancement decays over a relatively long time period; therefore, most of the enhancement is retained as the FID is collected.

b) The opposite result is obtained if the decoupler is not switched on until *the very instant* the excitation pulse is transmitted. Once the FID data have been collected, the decoupler is switched off until the next excitation pulse. This pulse sequence is called **inverse gated decoupling**. The effect of this pulse sequence is to provide a proton-decoupled spectrum with no NOE enhancement. Because the decoupler is switched off before the excitation pulse, the nuclear Overhauser enhancement is not allowed to develop. Proton decoupling is provided since the decoupler is switched on during the excitation pulse and the acquisition time.

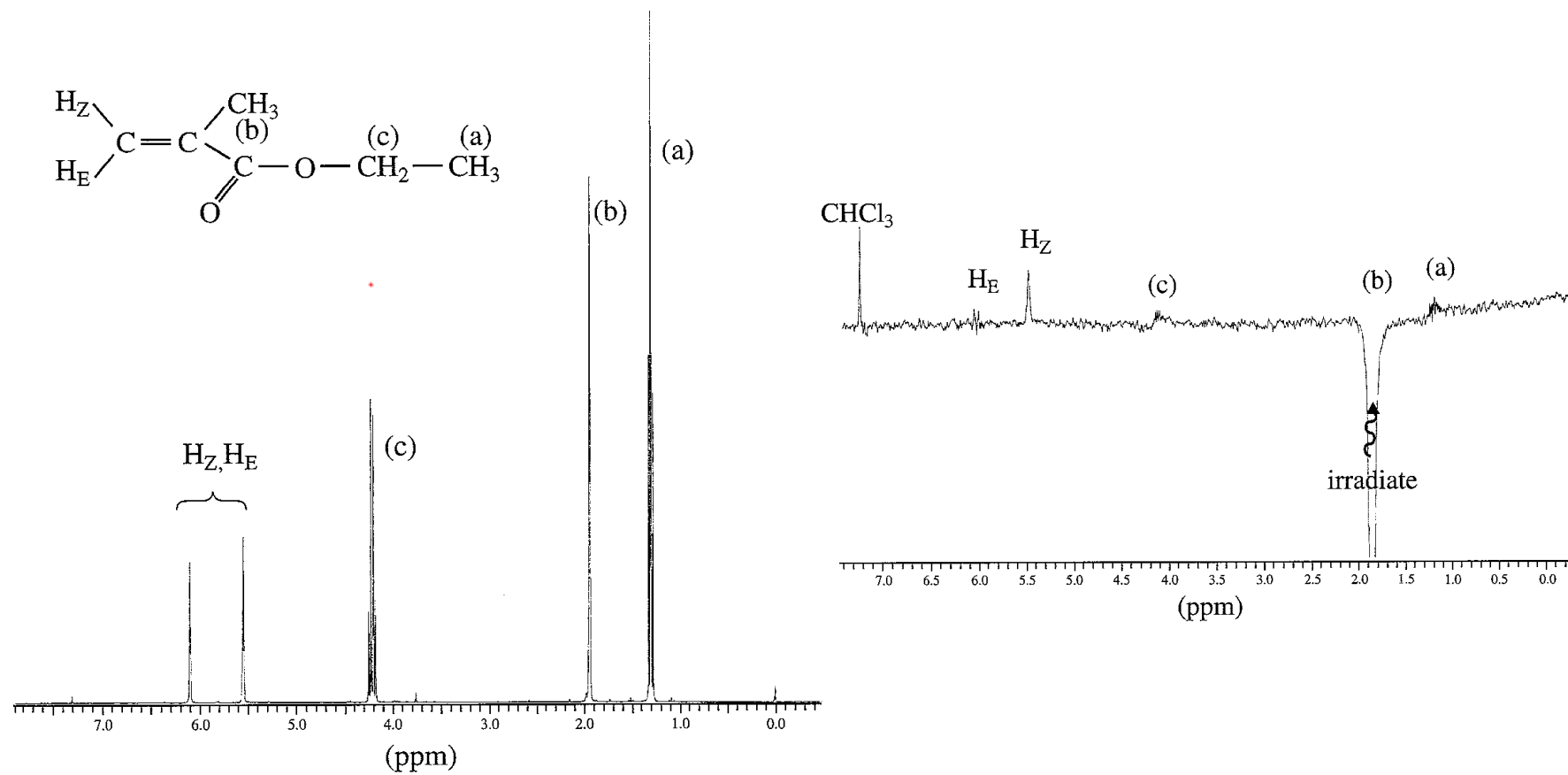
Nuclear Overhauser Effect (NOE)

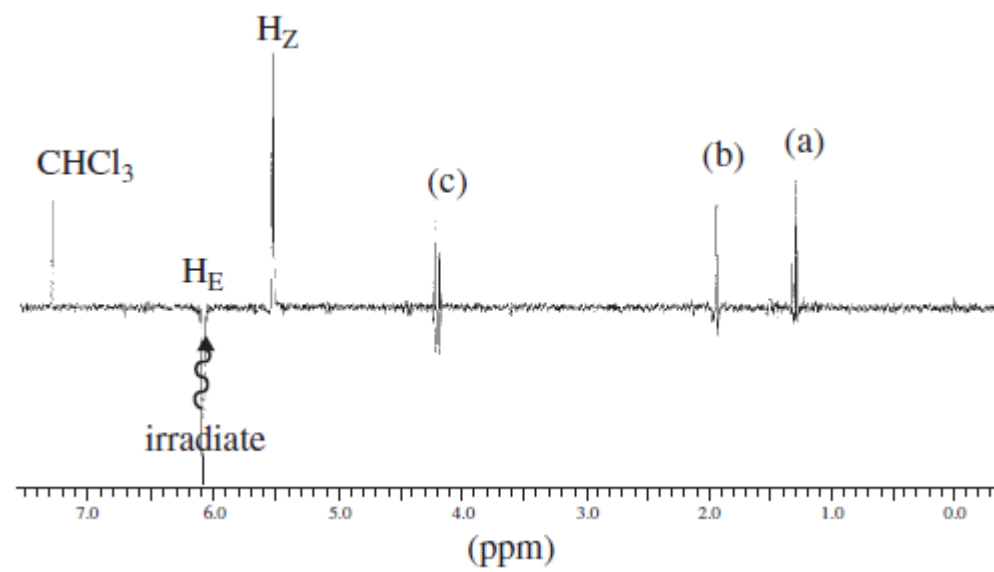
- NOE: Change in the intensity of a nucleus upon irradiation of a nearby nucleus

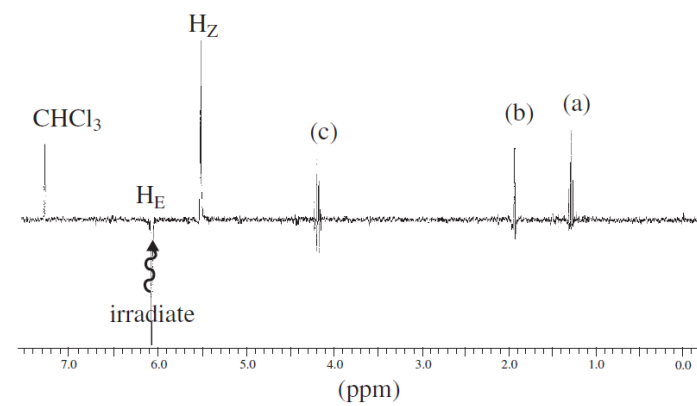
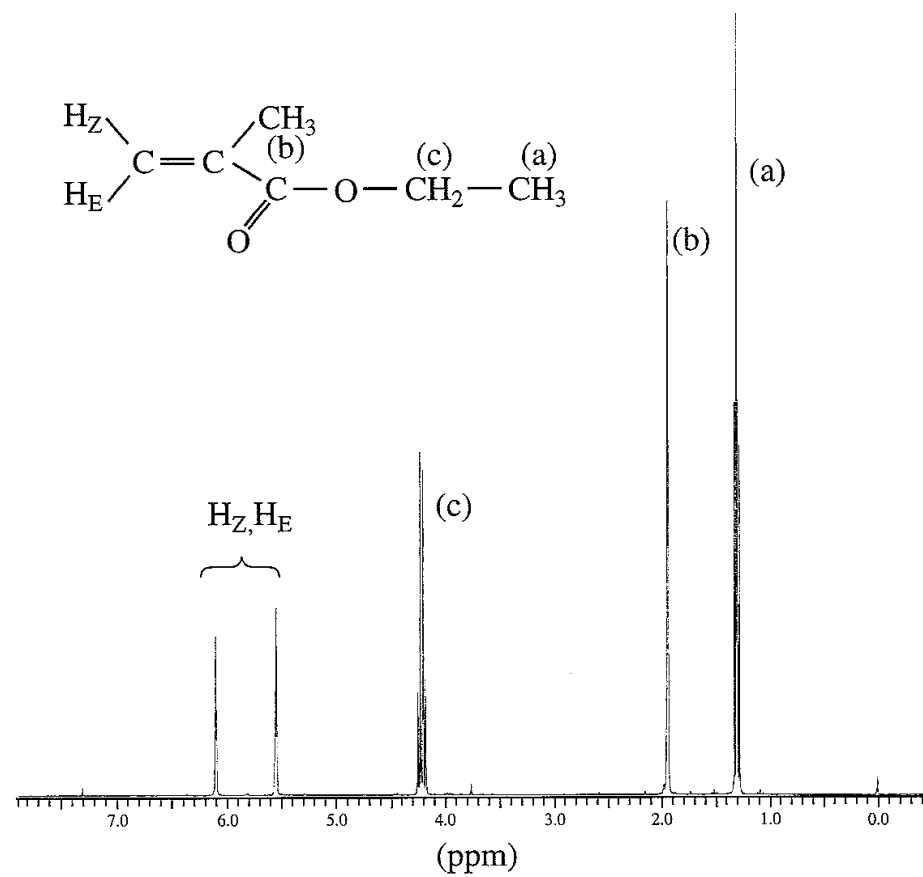


- Homonuclear NOE
- Distance dependent: **weak NOE** $> 3.5 \text{ \AA}$

NOE Difference Spectra

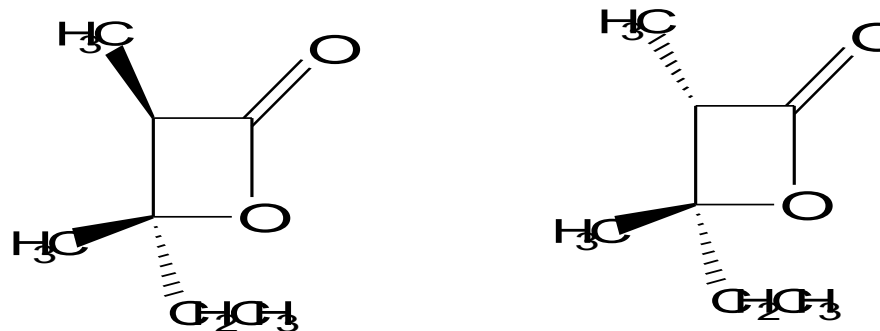




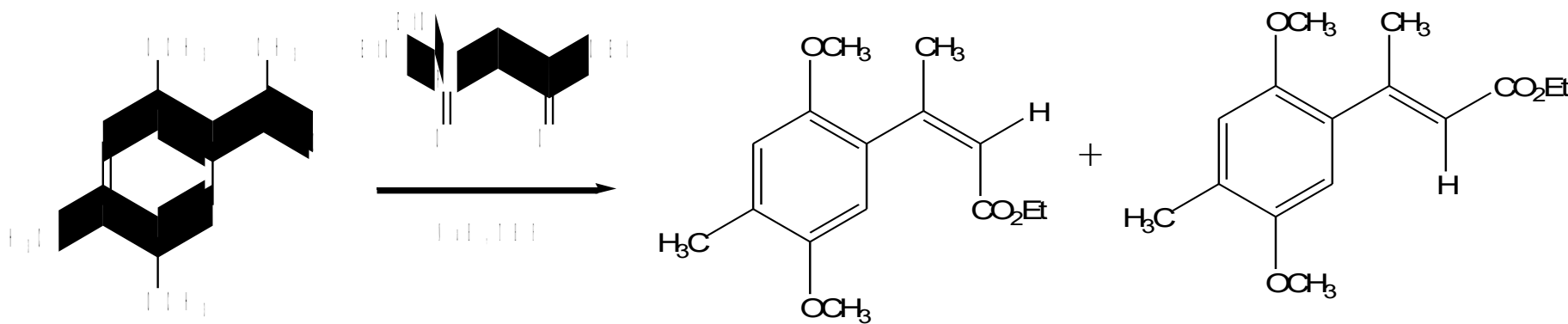


NOE in Stereochemistry

- How to distinguish between the following two diastereomers?

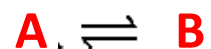


- How do we distinguish the cis and trans isomer in the following Horner-Emmons reaction?



Dynamic NMR Spectroscopy

- We use NMR spectroscopy mainly to determine the chemical structure of molecules
- But it is also a powerful tool to observe/detect dynamic processes within a molecule (bond rotation, ring inversion and tautomerism) or between molecules

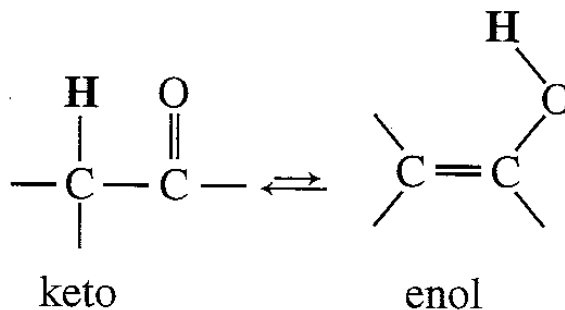


Slow exchange: rate of interconversion is slow on NMR time scale (typically ms), separate signals for A and B

Fast exchange: rate of interconversion is fast on NMR time scale, no separate signals for A and B

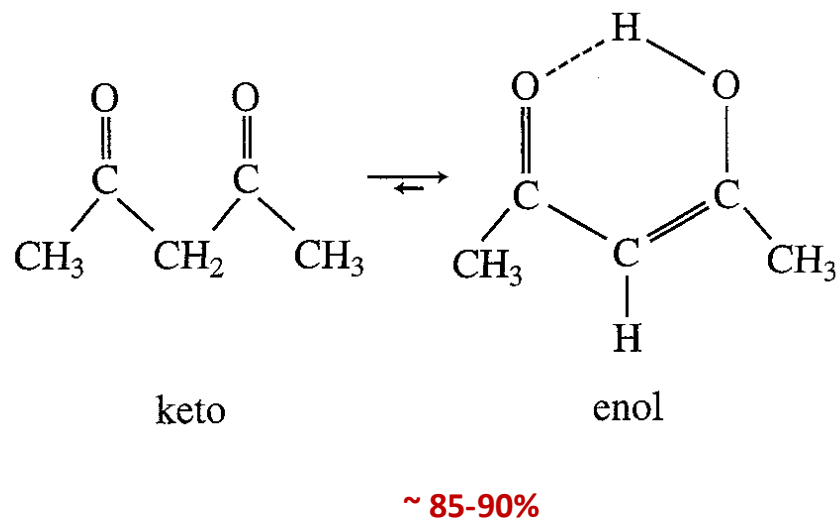
Tautomerism

Dynamic NMR: Studying processes that involve rapid interconversion between molecular species



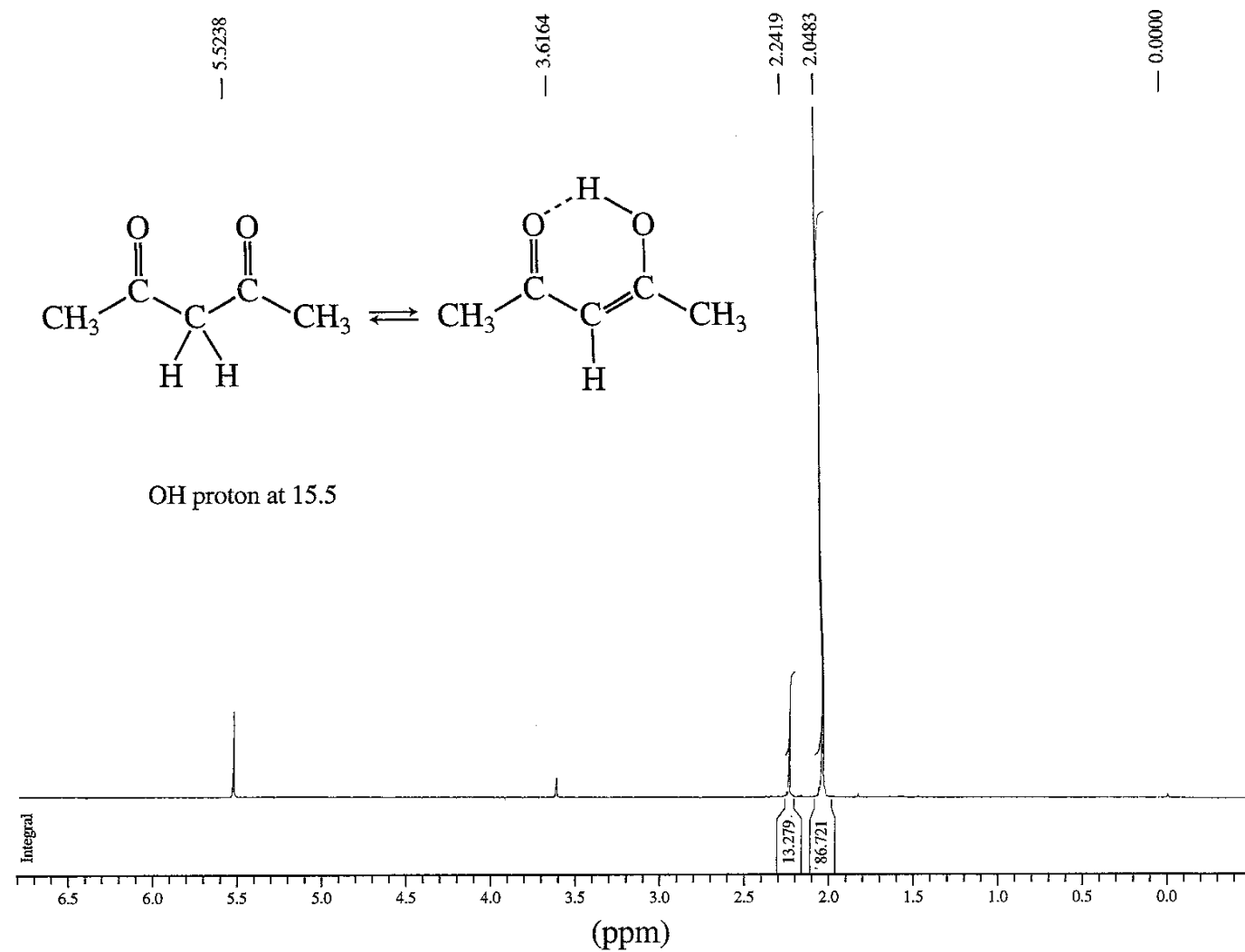
For acetone, much less than **1%** enol compared to the keto form in equilibrium

The situation is quite different in acetyl acetone



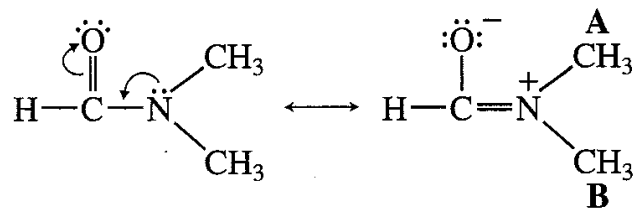
✓ The enol form is stabilized by

1. Delocalization through resonance
2. 6 membered intramolecular H bonding



- ✓ The keto-enol tautomerism is slower than the NMR time scale at RT
- ✓ What is the ratio of keto and enol form here?

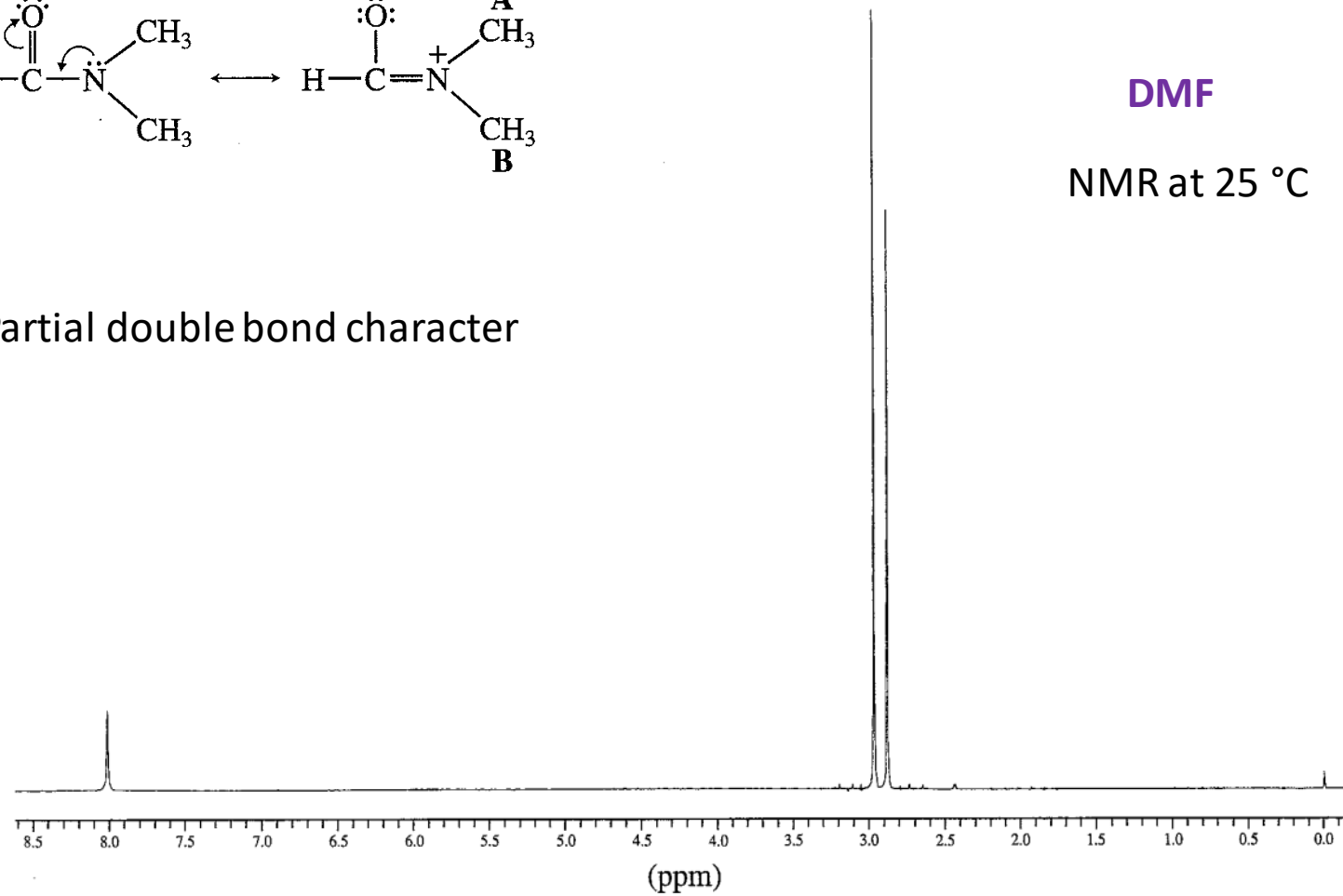
Amides

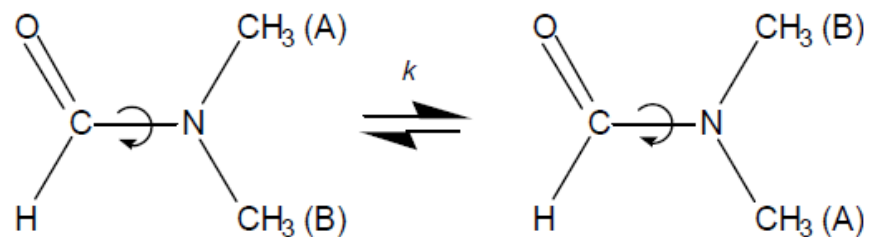


Partial double bond character

DMF

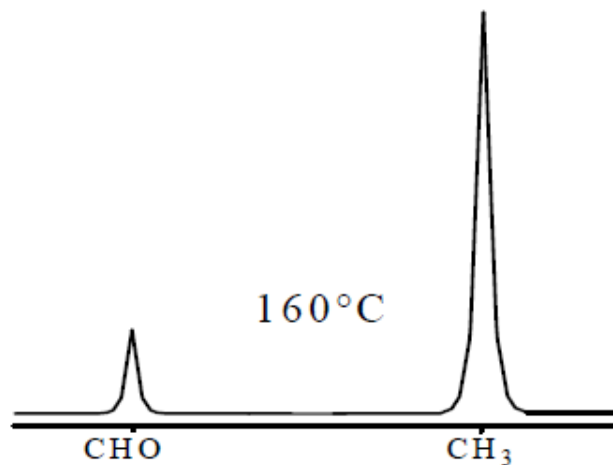
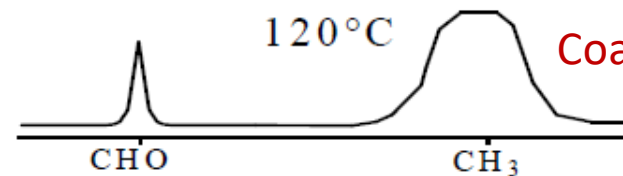
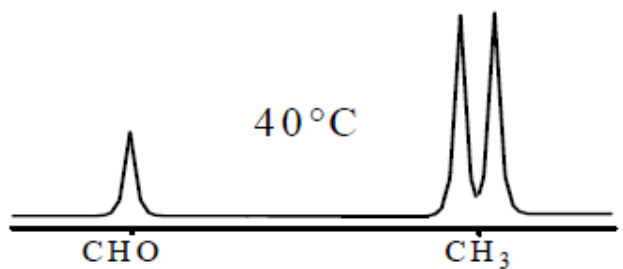
NMR at 25 °C





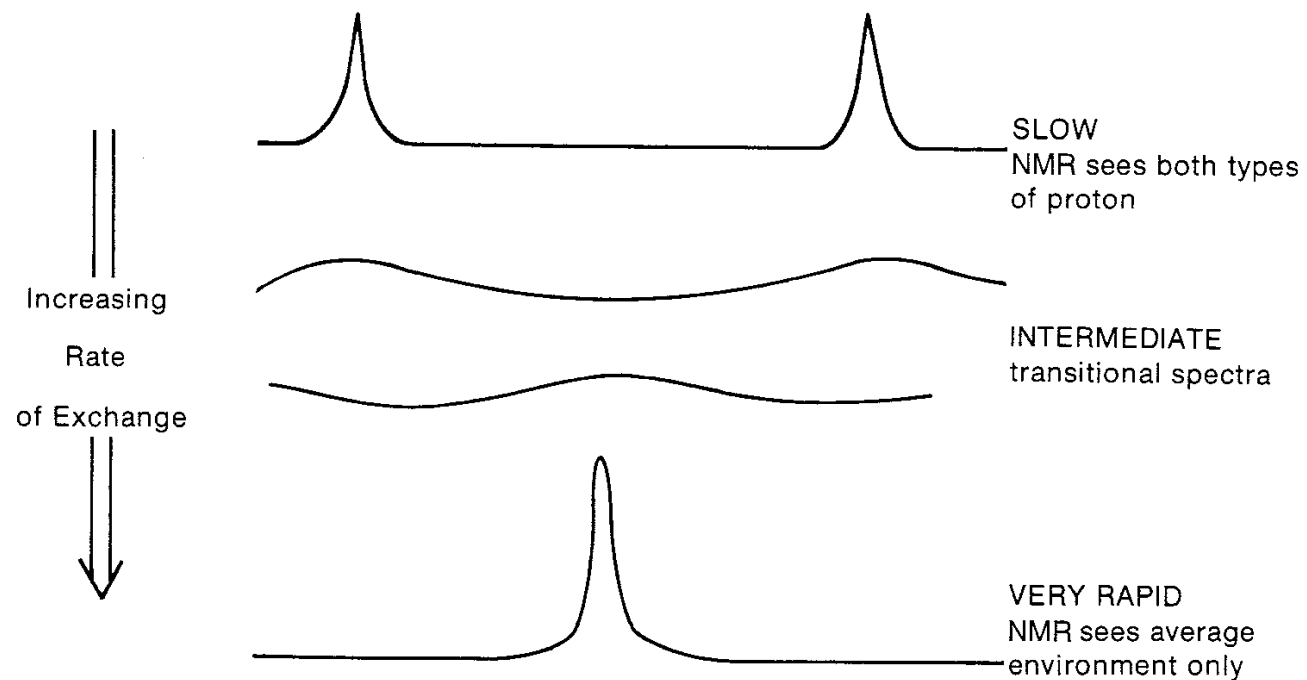
➤ Energy barrier of rotation: 88 kcal/mol

➤ Exchange is slow at 40°C



➤ The lifetime of the methyl groups in *cis* or *trans* position to the carbonyl group is relatively long near room temperature

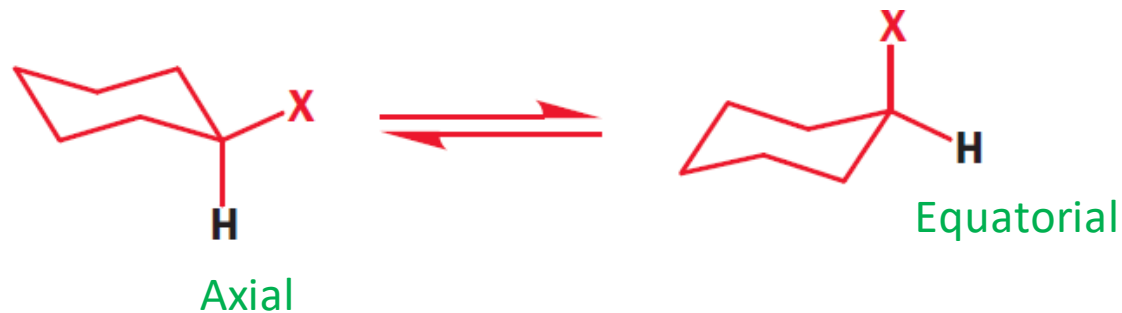
➤ The rate of exchange through bond rotation increases with temperature



- Beyond a certain temperature, the rate of exchange is so fast that the magnetic environments of the two sets of nuclei are identical and any possible distinction between the nuclei is lost
- Thus we have only one set of spins with a lifetime determined mainly by relaxation mechanisms

Cyclohexane

- The chair form has two types of Hs: Equatorial and Axial
- At room temperature, only one NMR signal is observed
- Rapid ring flipping



At room temperature, the NMR spectrum of cyclohexane shows only a single resonance peak. As the temperature of the sample is lowered, the sharp single peak broadens until at -66.7 °C it begins to split into two peaks, both broad. As the temperature is lowered further to -100 °C, each of the two broad bands begins to give a splitting patterns of its own.

Core DEPT Rules (quick recall)

- DEPT-90 → only CH
- DEPT-135 →
 - CH₃ → positive
 - CH → positive
 - CH₂ → negative
- Quaternary carbons → absent in all DEPT

Problem 1 Molecular formula: C₄H₁₀O

•¹³C NMR: 19, 31.7, 69.5 ppm

•DEPT-90: 31.7 ppm

•DEPT-135:

- +19 ppm
- -69.5 ppm

Predict structure